LESSON 2.7

Divide Decimals Using an Algorithm

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can divide with decimals by making the divisor a whole number first.

WHY THIS MATTERS A school bought new math workbooks as shown in the order form. What is the cost of each workbook?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Dividend *Spanish: Dividendo*	The total number being divided into equal groups (the number inside the division bracket).	In 84.6 ÷ 6, it is 84.6
Divisor *Spanish: Divisor*	The number of equal groups you are dividing into (the number outside the division bracket).	In 84.6 ÷ 6, it is 6
Quotient *Spanish: Cociente*	The answer when you divide.	84.6 ÷ 6 = 14.1
Decimal Division Algorithm *Spanish: Algoritmo de división decimal*	The steps for dividing decimals: slide the divisor's decimal point right until the divisor is a whole number, slide the dividend's point right the same number of places, then divide.	8.4 ÷ 2.1 → move the point in both → 84 ÷ 21 → divide, multiply, subtract, bring down → 4
Equivalent division *Spanish: División equivalente*	Multiplying both numbers by 10 or 100 gives the same answer.	7.56 ÷ 0.12 = 756 ÷ 12 = 63 (both multiplied by 100)
Annex a zero *Spanish: Agregar un cero*	Write a zero after the last decimal digit without changing the value.	7.2 → 7.20

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A rope 25.2 meters long is cut into 6 equal pieces. How long is each piece?

1. Write 25.2 ÷ 6. Put the quotient's point above the point in 25.2.
2. 6 goes into 25 four times, 1 left over. Bring down the 2 to make 12.
3. 12 ÷ 6 = 2. Check: 4.2 × 6 = 25.2.

Answer: 4.2 meters for each piece

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's do 8.4 ÷ 2.1. STEP 0 — MOVE THE POINT: how many places make 2.1 a whole number? (1 place, so 2.1 becomes 21.)

1. _____
2. _____
3. _____
4. _____
5. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The dividend is _____ and the divisor is _____ .
- I placed the decimal point _____ because _____ .
- When a remainder appeared I _____ .
- I checked my quotient by _____ .

Word bank dividend | divisor | quotient | decimal point | algorithm | annex | bring down | place value | estimate | check | multiply | subtract

WATCH OUT!

A common mistake

A common mistake in Divide Decimals is moving the decimal point in the divisor but forgetting to move it the same number of places in the dividend. For example, in 6.4 ÷ 0.8 a student changes 0.8 to 8 but leaves the dividend as 6.4, then divides 6.4 ÷ 8 = 0.8 instead of moving both points to get 64 ÷ 8 = 8. A second version of this mistake happens when the divisor has two decimal places, like 0.12: students move both points only 1 place instead of 2, turning 9.6 ÷ 0.12 into 96 ÷ 1.2 instead of the correct 960 ÷ 12. Count the divisor's decimal places first, then move BOTH points that same number of places.

LESSON 2.7

Divide Decimals Using an Algorithm

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Rewrite each problem by making the divisor a whole number, then solve.

Original	Multiply Both By	New Problem	Quotient
4.5 ÷ 0.9	10		
3.6 ÷ 0.4	10		

HINT The 'Multiply Both By' column is the key move: choose 10 or 100 based on how many decimal places the DIVISOR has.

SHOW YOUR WORK

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 14.4 ÷ 1.2?

(A) 0.12
(B) 1.2
(C) 12
(D) 120

HINT Dividing by a decimal is awkward — notice the divisor 1.2 has one decimal place. There's a move that makes it whole.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 3.78 ÷ 0.6?

(A) 0.063
(B) 0.63
(C) 6.3
(D) 63

HINT The divisor 0.6 has one decimal place — shifting both numbers one place makes the divisor a whole number.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is 8.4 ÷ 0.7?

(A) 0.12
(B) 1.2
(C) 12
(D) 120

HINT Ask yourself: how many groups of 0.7 fit inside 8.4? Since 0.7 is less than 1, expect MORE groups than 8.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is 15.6 ÷ 1.3?

(A) 1.2
(B) 12
(C) 13
(D) 120

HINT The divisor 1.3 isn't a whole number — shift both numbers before you divide anything.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Problem** Solve 6.4 ÷ 0.8.
2. **Move the divisor's decimal** 0.8 becomes 8 (moved 1 place right).
3. **Divide without moving the dividend** 6.4 ÷ 8 = 0.8

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** When you moved 0.8 to 8, what did you have to do to 6.4 as well?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does moving the point the same number of places in both numbers give you the same quotient — and why is it worth doing before you start the divide, multiply, subtract, bring down cycle?

SENTENCE STARTER Multiplying both numbers by ___ doesn't change the quotient because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.7

Divide Decimals Using an Algorithm

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Solve each decimal division problem.

Division Problem	Quotient
9.6 ÷ 0.8	
15.6 ÷ 1.3	
24.5 ÷ 3.5	
5.76 ÷ 0.24	
11.34 ÷ 0.9	

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each decimal division to its quotient.

_____ 7.2 ÷ 0.8
_____ 4.56 ÷ 0.6
_____ 13.2 ÷ 1.2
_____ 3.6 ÷ 0.12
_____ 10.5 ÷ 1.5
_____ 0.96 ÷ 0.8

(A) 9
(B) 7.6
(C) 11
(D) 30
(E) 7
(F) 1.2

3 **Write the name of the correct group on each line.**

Sort: should you multiply both numbers by 10 or by 100 to make the divisor a whole number?

Groups: **Multiply Both by 10** **Multiply Both by 100**

_____ 6.3 ÷ 0.7 → multiply by 10

_____ 8.4 ÷ 2.1 → multiply by 10

_____ 13.6 ÷ 1.7 → multiply by 10

_____ 7.56 ÷ 0.12 → multiply by 100

_____ 8.28 ÷ 0.36 → multiply by 100

_____ 9.45 ÷ 0.15 → multiply by 100

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A water tank holds 19.2 liters. It is poured into bottles that each hold 1.6 liters. How many bottles can be filled?

(A) 1.2 bottles

(B) 11 bottles

(C) 12 bottles

(D) 13 bottles

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Problem** Solve 9.6 ÷ 0.12.
2. **Count how far to move** 0.12 has 2 decimal places, so BOTH numbers move 2 places right. The divisor: 0.12 → 12.
3. **Move the dividend the same 2 places** 9.6 only has one digit after the point, so the student moved it just 1 place: 9.6 → 96. New problem: 96 ÷ 12 = 8.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does moving the point the same number of places in both numbers give you the same quotient — and why is it worth doing before you start the divide, multiply, subtract, bring down cycle?

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LESSON 2.7

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Challenge · Extension

MASTERY CHECK
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- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Decide whether each pair balances. Check Yes or No, then explain.**

Verify each division by showing that quotient × divisor = dividend. Are the two sides balanced?

	Left side		Right side	Balanced?
1	9.6 ÷ 0.8	=?	12 × 0.8 = 9.6	☐ Yes ☐ No
2	7.56 ÷ 0.12	=?	63 × 0.12 = 7.56	☐ Yes ☐ No
3	5.4 ÷ 0.9	=?	5 × 0.9 = 4.5	☐ Yes ☐ No

EXPLAIN YOUR CHOICE

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find Derek's Mistake

1. **Problem** 7.2 ÷ 0.45
2. **Multiply divisor by 10** 0.45 × 10 = 4.5
3. **Multiply dividend by 10** 7.2 × 10 = 72
4. **Divide** 72 ÷ 4.5 = 16

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A gas station sells fuel at $3.45 per gallon. A customer pays $27.60 total. How many gallons did the customer buy? Set up the decimal division, show how to convert it to whole-number division, and solve.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Why does moving the point the same number of places in both numbers give you the same quotient — and why is it worth doing before you start the divide, multiply, subtract, bring down cycle?

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